

Project Proposal: Applied Human Computer Interaction (AHCI)

Group Members

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1. Project Title

Smart FYP Management System

2. Main Features of the Product

The system is designed to digitize and streamline the complex lifecycle of Final Year Projects (FYP) at FAST NUCES. The key features are categorized by the interactions available to each specific user role:

For Students:

- **Proposal Dashboard:** A unified interface for submitting project ideas, managing team member details, and tracking real-time approval status from the department.
- **Milestone Submission Portal:** A dedicated area to upload deliverables (PDFs, Zip files) for specific deadlines (e.g., SRS, Mid-Term Defense) without navigating complex email threads.
- **Feedback & Grade View:** A centralized view for accessing supervisor comments and grades on specific deliverables, ensuring transparent communication.

For Supervisors:

- **Supervision Request Management:** An interface to review, accept, or reject incoming supervision requests from student groups.
- **Digital Rubric & Evaluation:** A structured grading interface allowing supervisors to input scores and text-based feedback using pre-defined digital rubrics, replacing manual spreadsheet entries.

- **Milestone Management:** Tools to set and broadcast specific deadlines (e.g., "Submit SRS by Friday") to their supervised groups.

For Jury/Panel Members:

- **Defense Dashboard:** A view of assigned student groups for evaluation during defense sessions.
- **Deliverable Review:** Direct access to review submitted documentation (SRS, SDS) and codebases (via GitHub links) within the portal.
- **Real-Time Grading:** The ability to input scores and feedback instantly during defense sessions using digital rubrics.

For Administrators:

- **Panel Assignment:** A drag-and-drop style interface (conceptual) to assign jury panels to specific student groups.
- **Global Announcements:** A broadcasting feature for updates regarding jury dates, template changes, or policy updates.

3. Expected User Groups

The product targets four distinct user categories within the university ecosystem:

1. **Students (Undergraduate):**
 - **Characteristics:** Tech-savvy but often overwhelmed by administrative deadlines. They require a system that reduces anxiety by providing clear status updates and submission confirmations.
 - **Goal:** To submit proposals and deliverables easily and receive timely feedback.
2. **Supervisors (Faculty Members):**
 - **Characteristics:** Busy professionals managing multiple groups simultaneously. They value efficiency and centralized information over complex navigation.
 - **Goal:** To track student progress, provide feedback, and grade submissions without managing disjointed email threads.
3. **Jury/Panel Members (Internal/External Experts):**
 - **Characteristics:** Evaluators who often have limited time during defense sessions. They need immediate access to project documents and grading tools.

- **Goal:** To review code quality and documentation quickly and submit evaluations in real-time.

4. Administrators/Coordinators:

- **Characteristics:** Staff responsible for the smooth operation of the FYP lifecycle. They need high-level oversight and management tools.
- **Goal:** To manage user registrations, assign panels, and ensure the entire batch complies with university timelines.

4. Domain Knowledge

Industry Context: The domain focuses on **Academic Administration and Workflow Automation** within higher education institutions, specifically tailored to the curriculum at FAST NUCES.

Current Landscape & Trends:

- **Digital Transformation:** Universities are moving away from paper-based or manual ad-hoc processes toward integrated digital ecosystems.
- **Lifecycle Management:** Unlike standard assignments, an FYP is a year-long lifecycle requiring distinct phases: Proposal Defense, Documentation (SRS/SDS), Development, and Jury Evaluations.
- **Process Bottlenecks:** The current domain is plagued by "disjointed communication channels" such as email threads and spreadsheets, leading to inefficiency and data inconsistency.
- **Technical Standards:** The domain requires adherence to modern development practices, such as Hexagonal Architecture, to ensure the system is scalable and maintainable for future academic batches.

5. Competitive Analysis

The Smart FYP Management System competes with both manual methods and generic educational tools.

Competitor Type	Product Examples	Strengths	Weaknesses	Differentiator of Our Product
Manual Systems	Email, Excel,	• Zero cost.	• Disjointed: Data is	Centralization

(Current Method)	Google Drive	• Everyone already knows how to use them. • Highly flexible.	scattered across different files. • No Tracking: Hard to monitor version history or status. • Inefficient: Grading requires manual compilation.	: Our system unifies proposal handling, milestone tracking, and grading into one platform, eliminating "email fatigue".
Generic LMS	Google Classroom, Canvas, Moodle	• Robust features for standard assignments. • Mobile apps often available. • Stable infrastructure.	• Lack of Specific Workflow: Cannot handle the specific "Proposal -> Defense -> Panel Review" lifecycle. • Generic Grading: Does not support complex, multi-role jury panels or specific FYP rubrics well.	Custom Workflow: Our system is specifically designed for the FAST NUCES FYP lifecycle (Proposal, Defense, Jury), which generic LMS tools cannot model accurately. +1
Project Management Tools	Trello, Jira, Asana	• Excellent for task tracking. • Great visual boards (Kanban).	• No Academic Context: No grading/rubric capabilities. • Complex for	Role-Based UX: We provide distinct, simplified interfaces for

			Faculty: Often too feature-rich/complex for non-technical supervisors.	Students, Supervisors, and Panels, rather than a generic project board.
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